

SAVITRIBAI PHULE PUNE UNIVERSITY

CROP ANALYTICS USING MACHINE LEARNING

A Dissertation

Under the Guidance of

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By

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Certified that this project report entitled:

Crop Analytics Using Machine Learning

is the bonafide work of KIRTIRAJ KADAM (CI18M16), a candidate for the degree of Master of Technology, who carried out the project work under my supervision.

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I would like to dedicate this work to my family members, especially, my Dad and my maternal Grandfather, Mr. Raghunath K Wadkar(Progressive Farmer - Kolhapur), also my friends who supported me throughout this process.

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-Kirtiraj Kadam

CONDUCTION CERTIFICATE



This is to certify that the project entitled:

Crop Analytics using Machine Learning

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We have no objection with he selecting Crop Analytics using Machine Learning as his project for the requirements of his M.Tech. He will be allowed to present the necessary technical details to the panel of examiners for the purpose of evaluation.

Date

Dr. V.C.V. RAO

Abstract

Crop classification and yield estimation has become an important activity in a country where agriculture is practised on a large scale. Identifying different types of crops using satellite imagery can help to formulate better yield output in the field of agriculture. Sentinel 2A data is high resolution satellite image data comprising 13 optical bands (with a resolution varying from 10 to 60 meters) and can be used to identify and classify different types of crops. The project work uses machine learning techniques like Random Forests, Support Vector Machines and Nave-Bayes classifier to classify different crops using Sentinel 2A data. After obtaining the ground truth data concerning different land cover classes, different classifiers have been trained. The trained classifiers have been used to perform the classification task and calculate the classification accuracies. The system will help in automatically identifying different crop patterns and facilitate calculating their yields. Also, the overall classification accuracy can sometimes be misleading, especially when the training samples are not uniform across the classes. Using uniform sampling the classification accuracy across all the classes has been observed to be uniform together with less inter-class mixing of pixels during prediction.

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Chapter 1

Introduction

Crop classification and yield estimation have become an important research activity in the field of agriculture as it can provide important information about the crops cultivated and harvested in a particular region. By studying different crop patterns the agricultural researchers can get an idea about the maximum estimate of different crops that are cultivated in a particular region and derive vital information that would be helpful in performing various decision making problems for managing various agricultural resources. Crop classification makes use of remote sensing data.

The image data needed to perform crop classification and yield is typically downloaded from freely available sources such as MODIS and Landsat. However, MODIS data is typically characterised by a mix-pixel problem (pixel that represents two or more entities on the ground) because of its low spatial resolution (250-500 m). Hence, better results can be achieved using a 30-meter resolution Landsat data, specifically, for the regions characterized by small agricultural fields. Earlier crop identification was performed by using optical remote sensing data but the accuracy for a given region usually depended on the noise and cloud free pixels, which hampered the identification of the crops during monsoon season.

Currently, Europe's Sentinel 2A data is being popularly used for crop identification and classification as its cameras can capture different colours of light (bands of electromagnetic spectrum) in different spatial resolutions, which can portray a different story about the agriculture environment. It has 13 spectral bands, comprising visible, near-infrared and

the shortwave infrared at different spatial resolutions ranging from 10 to 60 meters on the ground. Sentinel-2 can distinguish between dead vegetation (barren land) and live vegetation or between rich and poor crop yield as it has infrared footprints.

1.1 Challenges

Crop classification is one of the most challenging tasks among the land cover classification. The collection of training pixels for different crops using the ground truth data is one of the most important steps towards achieving desired accuracies for the crop classification process. Another challenge in crop classification is of acquiring cloud-free satellite images of a given crop area. Although Sentinel delivers images of each Earth location every six days, there is usually no guarantee of acquiring cloud-free images of a crop region over a specific period. For example, in India it is tricky to find cloud free images during the monsoon months. In such a situation SAR data can be used to collect complete time series satellite data over the region of interest for crop classification.

Another important aspect of crop classification and analytics is the use multi-temporal images (images of a crop region captured over different time periods) to observe and study the process of plant growth. These images can be a good data source for agricultural monitoring, and crop classification and mapping. However, processing of temporal images of a crop region requires more computational resources and downloading time over the internet.

Generally crop classification benchmarks developed for restricted areas and machine learning strategies checked on small test sites (around ten by 10 km) might not be scalable in several sensible cases. That is, what may fit for little test sites may not forever work for giant scale crop area. The standard machine learning techniques comparable to Thomas Bayes classifier, most chance methods, and Support Vector Machine perform well for small test sites, however is also computationally expensive for large scale problems, maybe at state level in India.

The concept of mixed pixels is another challenging aspect in crop classification. A mixed pixel represents an area occupied by more than one ground cover type such as an area at

edge of a large agricultural field. Mixed pixels also occur when objects imaged are relatively smaller compared to spatial resolution of the sensor, like long linear features (streams, village roads, field boundaries, etc.) and small objects (ponds, wells, bushes, etc.) in an agriculture land.

Finally, there could also be an overlap of pixels values across different crop classes during different stages of crop growth (crop signature overlap). The use of multi-temporal images may be used to resolve this issue to get better classification accuracy as different crops would have different responses to the spectral bands at different instances of time or crop growth.

1.2 Addressing the challenges

Crop classification revolves around the satellite image data, classifier models, and the ground truth. When conventional machine algorithms fails in dealing with big satellite data newer and faster techniques based on deep learning techniques can be explored. The ground truth data can be obtained from different experts to improve the classification accuracy.

When using multi-temporal data, cloud resources such as like Goggle Earth Engine or Amazon could be used to reduce the data downloading and processing time. Cloud platforms would enable scaling of solutions for smaller areas to relatively large agricultural tracts.

Due to fast technological changes, crop classification and mapping can still be relevant analysis discipline within the close to future for giant scale agricultural areas exploitation high resolution satellite imagery. The fusion of satellite technologies with high performance computations associate degreed AI would herald additional research challenges during this area.

1.3 Project Overview

The project work attempts to use machine learning algorithms to classify different crops using the sentinel 2A data and finding the crop patterns over a region. The system wwould

differentiate between different types of crops and also find other entities such as barren land, water bodies, urban sprawl, etc. A comparative study of different crop classification algorithms would be done and documented.

1.4 Importance of the Study

Crop classification is an important research activity for agricultural scientists. The use of machine learning algorithms to classify different crops has gained significant importance. Amongst these, ensemble learning methods such as random forests, gradient boosting, xgboost, etc. have gained wide popularity

Random forests carries with it an oversized individual call trees that operate as an ensemble. Every individual tree in the random forest spits out a category prediction and therefore the class with the foremost votes becomes our models prediction. Random decision forests correct for decision trees habit of overfitting to their training set.

In this context, applying *state-of-the-art* machine learning algorithms for crop classification becomes an important aspect of study and exploratory work. Crop classification would be helpful to agricultural insurance companies and fertilizer suppliers, who need to know different crops grown in their region and the areas of these crops.

1.5 Problem Statement

- Explore the use machine learning algorithms to automatically classify different crops using satellite images of an agricultural area
- Address the problem of imbalanced classification using uniform sampling
- Perform accuracy assessment of different machine learning algorithms used in the study

Chapter 2

Theory And Literature Review

2.1 Electromagnetic Spectrum

The ability of our eyes to see various objects is because of light which falls on objects and gets reflected on our retina and thus we can see objects. Human eye can only see visible light which can be seen in a rainbow (VIBGYOR) where Red has the longest wavelength while violet has the shortest so when all these colours are mixed we get white light, but light or visible light falls only in a small section of the electromagnetic spectrum there are waves which cannot be detected by the human eye like Gamma rays, X-rays, Infrared etc. These waves can help us to see beyond that of what a human eyes cannot see for example distant galaxies in space which are hidden behind dense interstellar clouds infrared telescopes can help us to find dim stars and can even monitor the temperature of planets in our solar system likewise here on Earth we can use these waves to study various fields like weather predictions, climate change, vegetation etc.

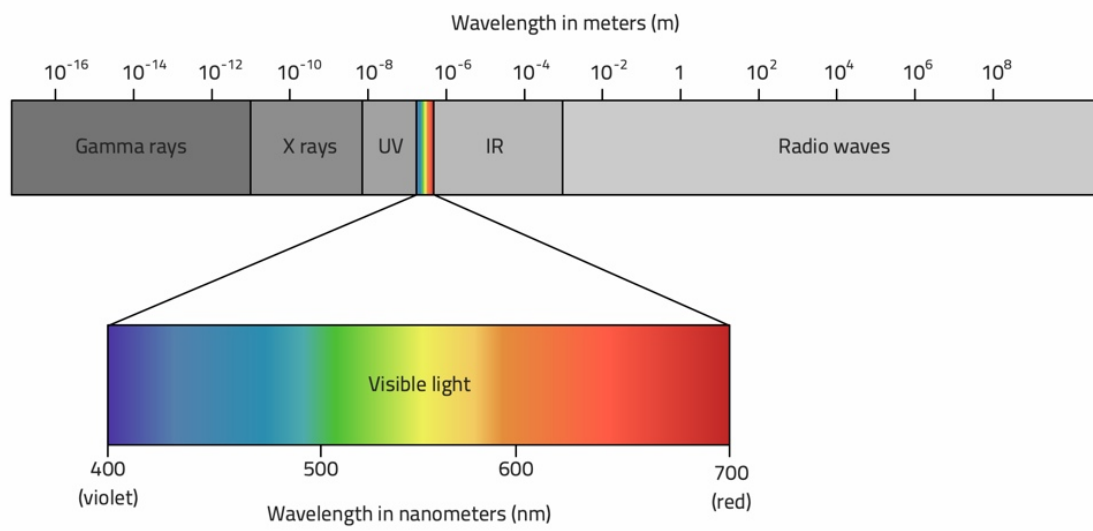


Figure 2.1: Electromagnetic Spectrum

2.1.1 Gamma Rays

Gamma rays are the most powerful rays in the electromagnetic and have shorter wavelengths that can pass through space they cannot be focused on to mirrors or lenses to create an image because they penetrate through them. The rays are detected from the gamma rays signal and converted into pairs of electrons and positrons. Due to their high power penetration they can damage bone marrow and internal organs and therefore they are hazardous and continuous exposure to them can kill humans. Nuclear reactions emit a lot of gamma rays also distant stars, galaxies emit gamma rays when they die which are then picked up by telescopes orbiting earth these radiations are then converted to create false colour images which we can see. As these rays have higher penetrating power they require large amount of shielding mass to reduce the levels that can cause harm to living things just like the x-rays gamma rays also get absorbed by materials which are having higher atomic numbers and high density materials.

2.1.2 X-Rays

The discovery of X-Rays were made by a physicist named Wilhelm Roentgen in the year 1895. X-Rays have higher energy than visible light but low energy than gamma rays these waves can penetrate through less dense materials like our skin, paper, cardboard which also have a low atomic number. X-Rays are widely used in the field of medical science as they can help us to see our internal bone structure by penetrating through the less dense material like our skin but a longer exposure to these rays can cause damage to our cells and tissues. X-rays cannot penetrate through dense material or material which are having higher atomic numbers like Gold, platinum even bones which are made up of calcium which have relatively higher atomic numbers so these materials absorb x-rays.

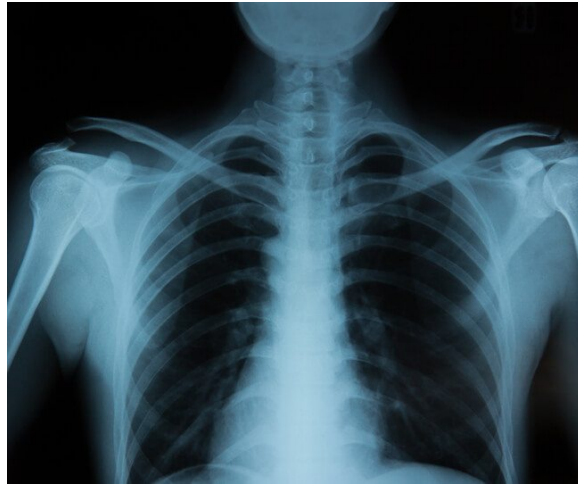


Figure 2.2: X-Rays

2.1.3 UV Rays

UV light is usually invisible light but it can be seen below 310 nm. There are 3 most common types of UV lights UV-A, UV-B, UV-C. UV-A is usually used for tanning, glow lights etc. UV-B lights are used in medical science like skin cure treatments, water purification, while UV-C is the most dangerous and can cause damage to eye retina and skin if exposed for a longer time. It can also kill germs, bacteria and dust mites as it penetrates through their membranes but it completely gets absorbed by the ozone layer so it never reaches Earth's surface. The sun is a source of energy on Earth as it emits visible light it also emits UV light most of it gets blocked due to the Earth's atmosphere but some of it does get penetrated.

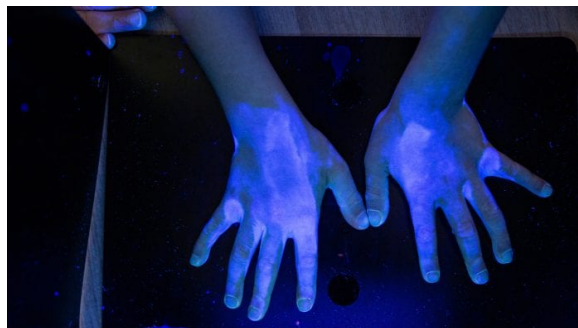


Figure 2.3: UV-Rays: Hands covered with germs can be seen under the light

2.1.4 Visible Light

The Visible light spectrum is the only segment which the human eyes can see i.e. our eyes can detect light which falls in the range 380 to 700 nanometres. The whole of electromagnetic radiation is light but our eyes can detect only a small portion of it which is the visible light. As we all know when the seven colours from the rainbow are mixed together it creates a white light, or if we project the white light through a prism we will get the seven lights which contain in the rainbow on the other side or vice versa. As the other radiations of the electromagnetic spectrum have their own wavelengths similarly each colour in the visible light has its own wavelength, like violet for example has the shortest wavelength (380 nm) compared to red which has the longest of all at 700 nm. The human eye consists of cone-shaped cells which act as receptors, the message is then transmitted to the brain which in turn produces the familiar sensations of colour. The objects reflect that part of light for example a Banana appears yellow to us that means it absorbs all the colours of light except yellow which is reflected, this reflected light falls on our eyes and thus we can see the fruit in yellow colour. If you take the same fruit deep into the ocean where there is less light the apparent colour would be different.

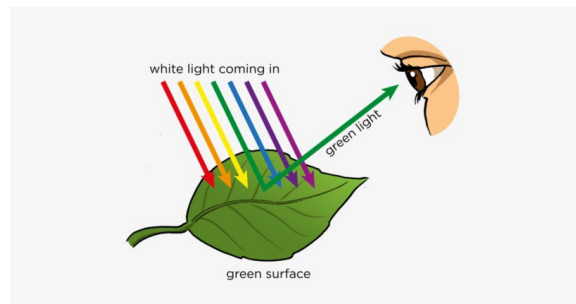


Figure 2.4: Visible Light

2.1.5 Infrared waves

Infrared radiation can help us to study vegetation on Earth it can penetrate through dense smoke, clouds and dust. By using these infrared radiation we can see much more clear story of our world even when there is no sunlight or any other source of light. Warm blooded animals maintain their body temperature so when infrared radiation is emitted they glow

even when it is dark. A good example of infrared is an infrared thermometer which was used to detect temperature of persons arriving and departing from an airport during Covid-19 pandemic, similarly different vegetation will reflect different thermal waves some may absorb while some may reflect all of them this will help us to categorize and also study various aspects of the crops.

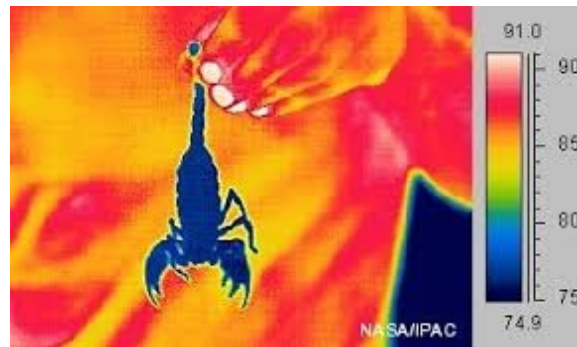


Figure 2.5: Thermal image of Warm vs Cold blooded

2.1.6 Radio Waves

Radio waves lie at the end of the Electromagnetic spectrum having frequencies ranging from 300Ghz to 3kHz. In vaccum the speed of radio waves is same as the speed of the light but in Earth's atmosphere they travel at a slightly lower speed. There are two types of Radio waves one are naturally occurring while the other are manmade which are mostly used in mobile radio communication, broadcasting, radar and GPS related communication whereas in nature the radio waves are produced by lightning and astronomical objects like pulsar stars which emit light in multiple wavelengths due to their high speed rotation and magnetic property. Just like Gamma and X-Rays Radio waves can penetrate through objects but the penetration power is low and hence they cannot penetrate through heavy metals like steel, iron.

2.2 Remote Sensing

Remote sensing refers to the detection or monitoring of physical characteristics of distant objects by sensing and capturing electromagnetic radiation reflected by these objects. Various sensors are used to capture the reflected electromagnetic radiation in different bands. In field of Remote sensing, the sensors can be of two types, active and passive. Active sensors send radiations towards the targeted objects and observe response to these radiations, while passive sensors collect radiation reflected from the objects where the source of light is usually the sun. In recent years many satellites have been launched in the outer space that collects information about different entities on the ground. The most popular of these satellites include the Landsat, MODIS, and Sentinel. These sensors on these satellites provide information (about different objects on the ground) in terms of images in various spectral bands such as Blue, Green, Red, NIR (near infrared), Panchromatic, etc. In the agriculture domain these bands are used for performing calculations for certain indices such as water index, vegetation index, NDVI, etc. in images of an agricultural area to predict crop pattern, crop yield, and cultivated area, amongst several others.

2.2.1 Satellite Data

A satellite image comprises reflectance values in several bands and is characterised by following resolution types:

- **Spatial Resolution:** Represents number of pixels used to obtain an image. For example, if a sensor has a resolution of 10 meters in certain band then every pixel in the image of that band would cover or represent an area of 10 m x 10m on ground.
- **Spectral resolution:** Specifies the number of different spectral bands that can be captured by the sensors of a satellite.
- **Temporal resolution:** It is the frequency of satellite visiting the same area of interest, for example, if a satellite passes over an area every 3 days then the temporal resolution would be 3 (image captured after every 3rd day).
- **Radiometric resolution:** It is the ability to detect small differences in electromagnetic

radiations. In satellite images the radiometric resolution denotes how detailed each pixel is and is indicated by units of bits. For example, a 8-bit resolution would have 256 gray levels, varying from 0 to 255.

2.3 Image Composites

Image composites are made up of two or more photographs stacked and flattened to create one image. Composites are used for various purposes like removing the noise from a low light photo, duplication of objects in the image for filmography, reduction of overexposed portions in the image, creating multispectral images etc. In the field of remote sensing each band of the satellites may be displayed in one grey scale image at a time or in combination of three or more bands as a colour composite image. We have three different composites used in remote sensing, one is the false colouring method while the other one is a natural composite and the third one is monochrome of three or more bands to create a greyscale image.

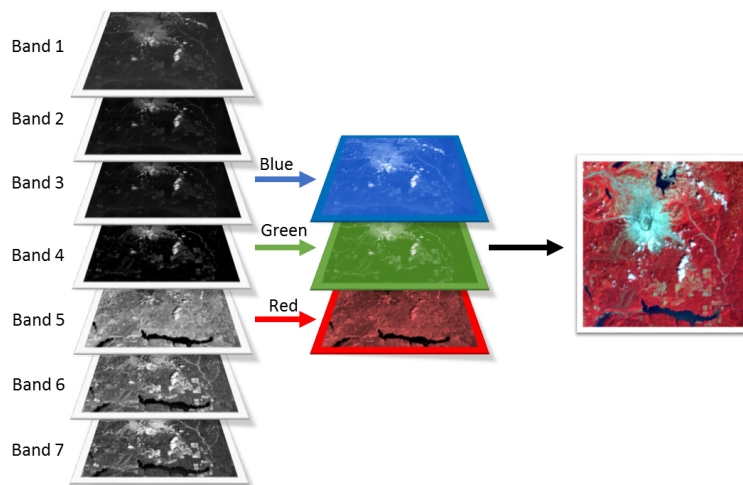


Figure 2.6: Image Composite: Stacked to create one image

2.3.1 False Colour Composite

As we know the electromagnetic spectrum consists of several other radiations like UV, X-Rays, IRs etc. these radiations cannot be seen by the human eye. When data is acquired using these radiations we can get a much more clearer data but to represent this data into an image form we use a technique called as false colouring where the visible as well as non-visible parts of the data are false coloured to create a multispectral image representation of the data. False colouring allows us to visualize the wavelengths of light that cannot be seen by the human eye. This technique is also used in astronomy where deep sky objects which don't reflect light if seen from an ordinary telescope hence they need specially modified cameras which use infrared light to sense the radiations coming from deep sky galaxies or objects. There are different false colour compositions to differentiate between various functions

1. Near infrared as red, green as blue, red as green. This type of combination is used in fields which are related to agriculture and vegetation health.
2. Shortwave infrared as red, near infrared as green, and green as blue, mostly used to depict floods or burned land.
3. Blue as red, two different infrared bands as green and blue. Used to classify between snow and clouds.
4. Thermal infrared used as grey shades to depict the temperature.

Here we are going to take the Near infrared band as Red, Green band as Blue while Red band as Green so our combination would look like NIR $\rightarrow$ Red, Green $\rightarrow$ Blue, Red $\rightarrow$ Green. Near infrared, red, green: Plants and Vegetation absorb red and reflect near infrared and green light thus areas which are plant covered appear red in our false composite. Denser plant growth show darker shades of red on the image. Barren land and cities appear grey and clear water appears black as it absorbs all three wavelengths, in our image the some parts of the water is not clear i.e. materials are present in the water reflect light and therefore the water appears blue.

2.3.2 True Colour Composite

True colour image as the name says is an image which depicts the real colours where the Red, green and blue bands correspond to the RGB channel on the computer screen. The image produced would be a real one as how we would see from our eyes, vegetation appears green, water to be blue and the clouds are white. The image taken from a DSLR or mobile uses a CCD or CMOS sensor which captures the light from surrounding objects and creates an image is a true colour or natural image whereas a satellite image consists of different wavelengths that can be visible and non-visible to the human eyes.

2.4 Accuracy Assessment

2.4.1 Confusion Matrix

Confusion matrix is used to summarize the performance of a particular classification or algorithm by visualizing analytics such as precision, recall, specificity etc. They are useful as we get the true readings of true and false cases from a number of samples. Consider the example of Covid-19 Patients there are four various scenarios while taking a covid test.

Confusion Matrix		
	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Figure 2.7: Confusion Matrix

1. You are infected and the test shows YES. Then this is True Positive (TP)

2. You may not be infected but the test shows Yes. Then this is False Positive (FP)
3. You may not be infected and the test shows No then its True Negative (TN)
4. You may be infected and the test says No then its False Negative (FN)

2.4.2 Sensitivity

Sensitivity is the ratio of correctly labelled positive samples by the system from all the correct samples. In short we can say as to how many samples does our system predicts true values from the actual true values. For example, The system which predicts correct cases of covid positive patients from a group of all covid positive patients can be defined as sensitivity or recall

$$Sensitivity = TP/[TP + FN]$$

2.4.3 Specificity

Specificity deals with the number of correct negative samples detected by the system against the total negative samples present in the group

$$Specificity = TN/[TN + FP]$$

2.4.4 Accuracy

Accuracy is the ratio of correctly labelled samples against the pool of samples. That means the samples which are really Positive or Negatives, lets understand this with a help of an example. The number of real positive covid-19 patients with addition to the number of patients whose report were to be negative to the Sars-Covid-19 virus against the number of all the people who took the test. The equation is given by,

$$Accuracy = [TP + TN]/[TP + FP + FN + TN]$$

2.4.5 Prevalence

Prevalence is how often does our model predicts the true cases and is defined as TPN

2.4.6 Kappa factor

Kappa enables comparison of the performance of any classifier to the baseline of a random classifier. The value of κ is always 1 when accuracy is 1. Specifically,

$$\kappa = \frac{\text{Accuracy} - \text{Random Accuracy}}{1 - \text{Random Accuracy}}$$

, where

$$p1 = \frac{TP + FN}{TP + FP + TN + FN}$$

$$p2 = \frac{TP + FP}{TP + FP + TN + FN}$$

$$\text{Random Accuracy} = p1p2 + (1 - p1)(1 - p2)$$

2.4.7 Uniform sampling

Imbalanced classification occurs when there is unequal distribution of class samples across different classes in the dataset. In imbalanced classification the classification algorithms have a bias towards the majority classes resulting in higher misclassification in the minority classes. In order to avoid bias in classification accuracy due to skewness in number of available training samples in each class, we carry out uniform sampling of different crop categories for training and testing purpose. Uniform sampling ensures that the accuracy across all the class is uniform.

2.5 Machine learning Algorithms

2.5.1 Decision Tree

A decision tree lets you take decisions to get the desired outcomes by choosing the lowest path or resource cost. The tree consists of internal nodes which represent test on an attribute, the branches represents the outcome, leaf node represents a class label while the root to leaf represent rules which would help to classify. Decision trees are powerful and used in supervised machine learning algorithms.

2.5.2 Random Forest

Random forest (RF) is a supervised, flexible and is easy to use machine learning algorithm which can produce accurate results. It is used for classification and regression based problems. The algorithm can handle values which are missing and can create a high accuracy for missing data.

As the name says the algorithm generates forest with a group of decision trees, with the number of trees in the forest the predictions become more powerful and robust. For classification of a new object based on the attributes of each tree gives a classification result and a vote is taken for that class, the classification is done on the basis of number of votes gained by other trees for a particular class in the forests while in regression the average of the outputs by different trees is taken. Overfitting can make the results worse but in Random forest if there are enough trees it won't overfit the model.

Random forest is an ensemble method for classification, where response of several classifiers are combined to get final prediction result. It has two important parameters, namely, *n_tree* (number of trees to form ensemble) and *m_try* (number of variables/predictors used to split the nodes). The best split for a node increases the accuracy of the classification.

Random forest algorithm has several benefits in remote sensing applications:¹

- RF can handle large data sets and input variables

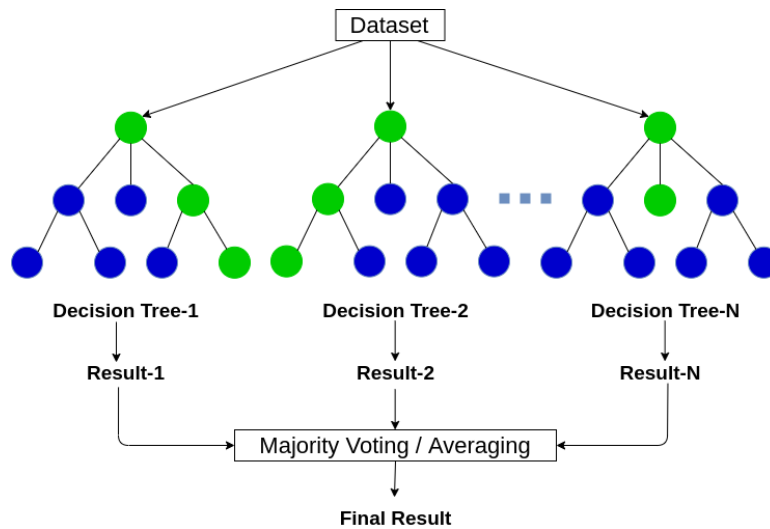


Figure 2.8: Random forests trees and classification/regression process

- Weak classifiers can pair up to become strong classes
- RF can be used for both classification and regression tasks
- RF is robust for outliers as well as to noise

2.5.3 Support Vector Machine (SVM)

SVM is a supervised learning model used in machine learning which makes use of statistics for classification of data. Classification task consists of training and testing data with some data constraints. A training set may consist of one target value and several attributes. The model is trained by feeding known labels to check whether its producing the desired output or not. The step in SVM segmentation involves identification as they are closely linked to known categories and is called as feature selection. Feature selection and SVM segmentation can be used even though the prediction of not known samples is not required. A support vector machine works by drawing a best hyperplane (in 2D it would be a simple line) which would separate two distinct groups of data samples so the objects which would fall to the one side of this line would fall in say suppose group a and the objects other side of the line would be categorized into group b.

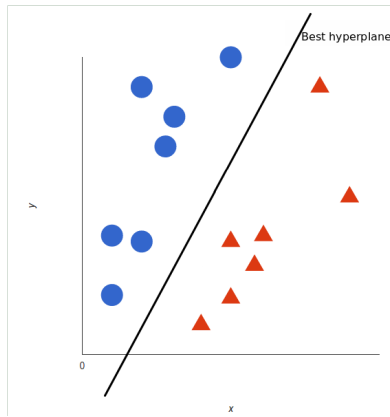


Figure 2.9: The best line drawn such that it classifies two distinct classes

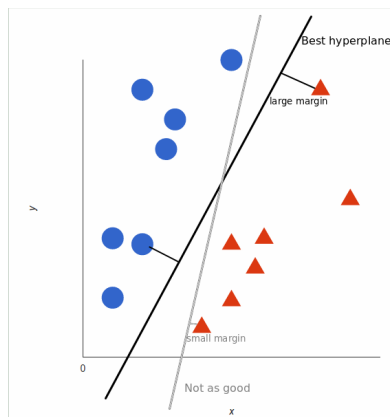


Figure 2.10: Not all lines are created equal

- SVM's can classify even if we have no idea about the data.
- Can classify unstructured and semi structured data like text, Images and trees.
- Risk of over-fitting is less in SVM.

2.5.4 Naive Bayes

Naive Bayes algorithm is primarily based totally on assumption of independence among the capabilities, that is, the values acquired for distinctive capabilities are unbiased of every different. The educated version may be constructed speedy and is pretty powerful in acting

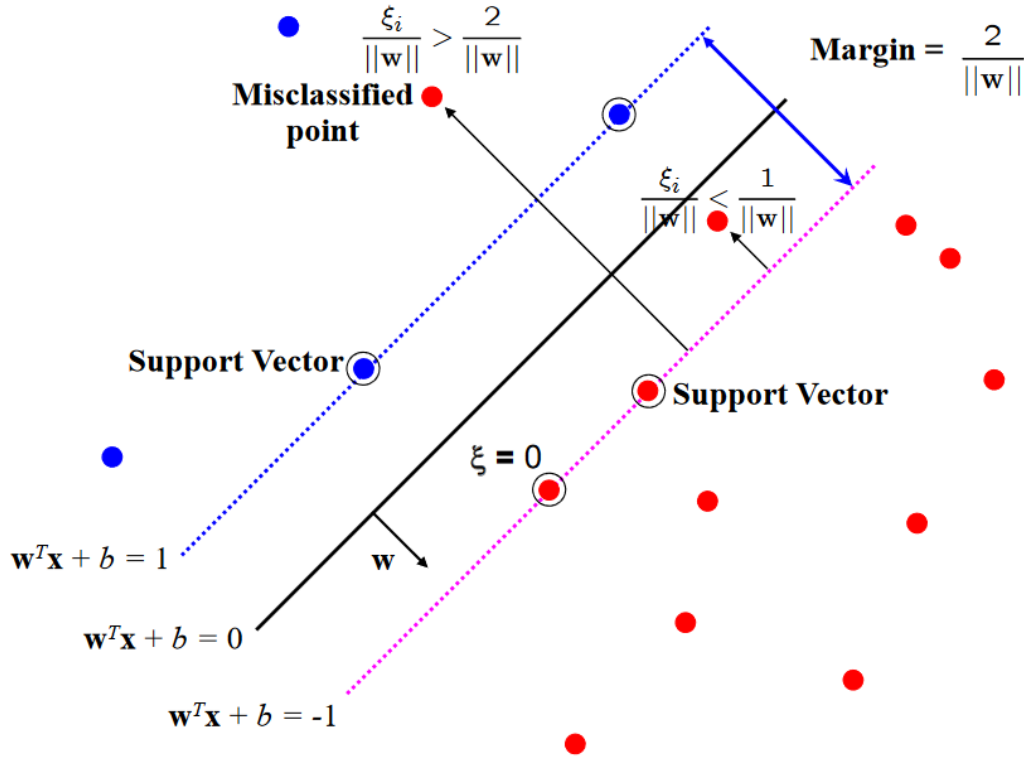


Figure 2.11: Separating hyperplane and support vectors in SVM

brief predictions. The set of rules is called naive as it makes the belief that the incidence of a selected characteristic is unbiased of the incidence of different capabilities.

For example, in case you need to perceive a fruit on its shape, colour and flavour then a yellow coloured C shaped fruit could be maximum in all likelihood a banana although those capabilities rely upon every different or at the presence of the alternative capabilities.

These properties subsequently make a contribution to the opportunity that the fruit is a banana and thats the motive its referred to as nave. The name Bayes refers back to the statistician and logician Thomas Bayes and the theory named after him is called Bayes theorem. This theorem offers a way to calculate the conditional opportunity, that is, the opportunity of an occasion primarily based totally on preceding expertise to be had at the events.

The Bayes classifier estimates possibilities of occurrences of various characteristic values for

distinctive training in a education set. It makes use of those possibilities to categorise do not forget pattern. If $\bar{A}$ is an array of M attributes $(A_1, A_2, \dots, A_M)$ comprising the training set, then $P(\bar{A})$ refers to the probability that a training pattern has the attribute array $\bar{A}$.

Assume that the training set has patterns from m classes $C_1, C_2, \dots, C_m$. If $P(C_k)$ is prior probability that a training pattern belongs to class C_k , $P(C_k|\bar{A})$ is posterior probability of attribute array $\bar{A}$ belongs to class C_k , $\bar{A}|P(C_k)$ is the conditional probability that a training patterns of class C_k has attribute array $\bar{A}$, then according Bayes' theorem

$$P(C_k|\bar{A}) = \frac{P(C_k)P(\bar{A}|C_k)}{P(\bar{A})} \quad (2.1)$$

To classify a pattern with attribute array $\bar{A}$, we estimate the probability $P(C_k|\bar{A})$ for each class and assign the pattern to class with highest such probability. Hence, we need to maximize $P(C_k|\bar{A})$, which is same as maximizing the right hand side of Eqn. 2.1. As denominator $P(\bar{A})$ is constant, we need to maximise only the numerator term $P(C_k)P(\bar{A}|C_k)$. However, estimating $P(\bar{A}|C_k)$ needs large training sets that consider all possible values for the attributes, So it is assumed that the attributes values are class conditionally independent, that is

$$P(\bar{A}|C_k) = \prod_{i=1}^M P(A_i|C_k) \quad (2.2)$$

After making the above assumption, the classifier is called as Naive Bayes classifier and is as given below

$$P(C_k|\bar{A}) \sim P(C_k) \prod_{i=1}^M P(A_i|C_k) \quad (2.3)$$

2.6 Literature survey

Orynbaikyzy et al.² have used fusion of Sentinel-2 and SAR data and Random forests to improve crop classification accuracy in comparison to using only either optical data or SAR

data. Liang et al.³ have used high-resolution, multi-spectral data together with Lidar data to derive spectral, textural and structural features for mapping target vegetation species using SVM and Random forests. Yi et al.⁴ have used multi-temporal Sentinel-2 data (10m spatial resolution) and Random forest algorithm to generate the crop classification map for the Shiyang River Basin. Their results demonstrated that the combination of multi-spectral and multi-temporal Sentinel-2 data improved the accuracy of crop classification significantly.

Neetu and S. S. Ray⁵ used numerous machine learning algorithms for crop classification using Sentinel-2A information where they filtered the image by removing clouds, True Color image, False Color image and NDVI computed image were used for distinctive crop features. Feature Collections were generated by choosing the coaching polygons primarily based upon the bottom truth collected using smartphone based automaton application. When training datasets generation totally different machine learning algorithms (CART (Classification And Regression Trees), RF (Random Forest), SVM (Support Vector Machine) were used for classifying the image victimisation GEE (Google Earth Engine) wherever they got Associate in Nursing accuracy of CART- 73.4 %, RF-93.3 percent and SVM -74.3 percent however in this technique the vegetable class were mixed with different classes.

R. Saini and S.K. Ghosh⁶ classified crops by making use of Sentinel-2 information that consisted on four bands (NIR, Red, Green, Blue) 10m resolutions on a 1049.31 square km land. The chosen space is split into the eleven Land Use Land cover (LULC) classes: High-density Forest, Low-Density Forest, Sandy area, Water, Fallow land, Built-up, Orchard, Wheat, Sugarcane, Fodder and different crops. the category named as Other crops consists of the double crop like Trees and Wheat, Mustard and Wheat, vegetable (Radish, Cabbage, Cauliflower). the foremost crop within the selected region is Wheat and Sugarcane. each classifiers (RF and SVM) were enforced using R programming language which is an open source language where they got an accuracy of 84.22 % by using Random Forest and 81.85 percent using SVM.

Jitendra Singh et al.⁷ used sentinel-1 artificial Aperture microwave radar (SAR) and sentinel-2 information two optical mental imagery in crop identification for an Indian region for crop classification. A multi-class classification algorithmic program supported random forest was applied to the options extracted from the satellite digital audiotapea sets conjointly a group of 10 Sentinel-2 optical images were acquired. These images were processed

to level 2A (i.e., surface reflectance values with masks for clouds, cloud shadows, snow and water) at 20m spatial resolution using sen2cor toolbox. They got an accuracy of 85%.

There are several other efforts in using freely available satellite data for crop classification and vegetation mapping. Zheng et al.⁸ and Waldner et al.⁹ have used MODIS and Landsat for vegetation mapping. The recently launched Sentinel-2 satellite has thirteen spectral bands with three different spatial resolutions. Its data has been used for various remote sensing applications (Whyte,¹⁰ Sonobe,¹¹ Korhonen,¹² Belgiu,¹³ Wang,¹⁴ Wang,¹⁵ Hawrylo¹⁶). Sonobe¹¹ have established that band 4 (Red) of Sentinel-2 and VV polarization data of Sentinel-1 are important for crop classification. Korhonen¹² has used Sentinel-2 data to estimate boreal forest canopy cover and leaf area index.

Zheng⁸ have used SVM model to identify different crop types using Landsat Normalized Difference Vegetation Index (NDVI) data. Shao¹⁷ has used neural networks, classification and regression trees and SVM for crop type mapping using MODIS NDVI data and has shown that SVM achieves better classification accuracy than the other two methods. Using an ensemble of five supervised methods and comparing it with extreme gradient boosting (XGBoost), SVM, logistic regression and multilayer perceptron for land cover classification using Landsat-8 data, Man¹⁸ showed that ensemble method gives highest accuracy. Son¹⁹ has used Sentinel-1A data to classify crops using Random forests and SVM model and has shown that Random forests classifier performed better than SVM classifier. Similarly, Whyte¹⁰ has shown that Random forests perform better than SVM for wetland mapping using Sentinel-1 and Sentinel-2 data. Li²⁰ has shown that Random forests and SVM are suitable for image analysis and classification in agricultural domain.

Chapter 3

Crop Classification

3.1 Study Area and Input Data

Data used in this project are images captured by Sentinel-2A and Sentinel-2B satellites, these satellites rolled out on June 22, 2015 & March 7, 2017 respectively to provide coverage of land surfaces. When these two satellites are operated simultaneously the temporal resolution obtained is 5 days i.e. the area of interest can be visited every 5 days this would help farmers and agricultural scientists to have a good grip on the crop fields and would help them to provide necessary information about the crops such as the crop health, its yield etc. Sentinel-2 has four bands at 10 meter resolution and several spectral resolutions, i.e. bandwidths with radiometric resolutions of 12 bits which means that $2^{12}=4096$ distinct values can be obtained to distinguish. Table 1 shows data about Sentinel-2 satellites 13 different spectral bands.

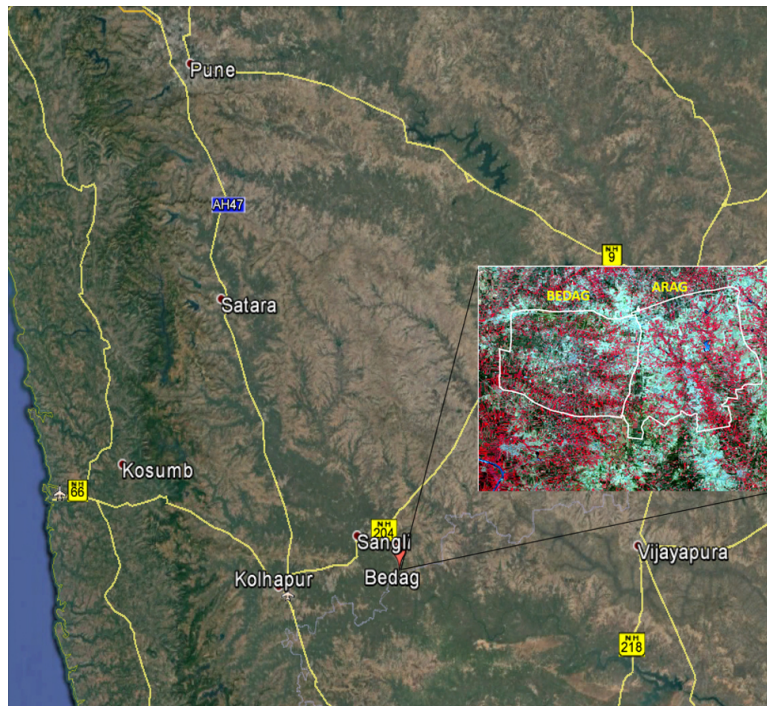


Figure 3.1: Study Area

3.2 System Architecture

The Data of crop images over a region can be downloaded from European satellite sentinel 2A and Landsat data then by using a trained sample classification of crops will be performed by the system.

The model will be trained using Random Forest (RF) and classified using SVM. There are three main components in this system the data will be gathered then from the gathered data model will trained to produce results these results will be matched with the ground truth i.e. actual ground truth will be checked and the results would be compared with the ground truth for the classification.

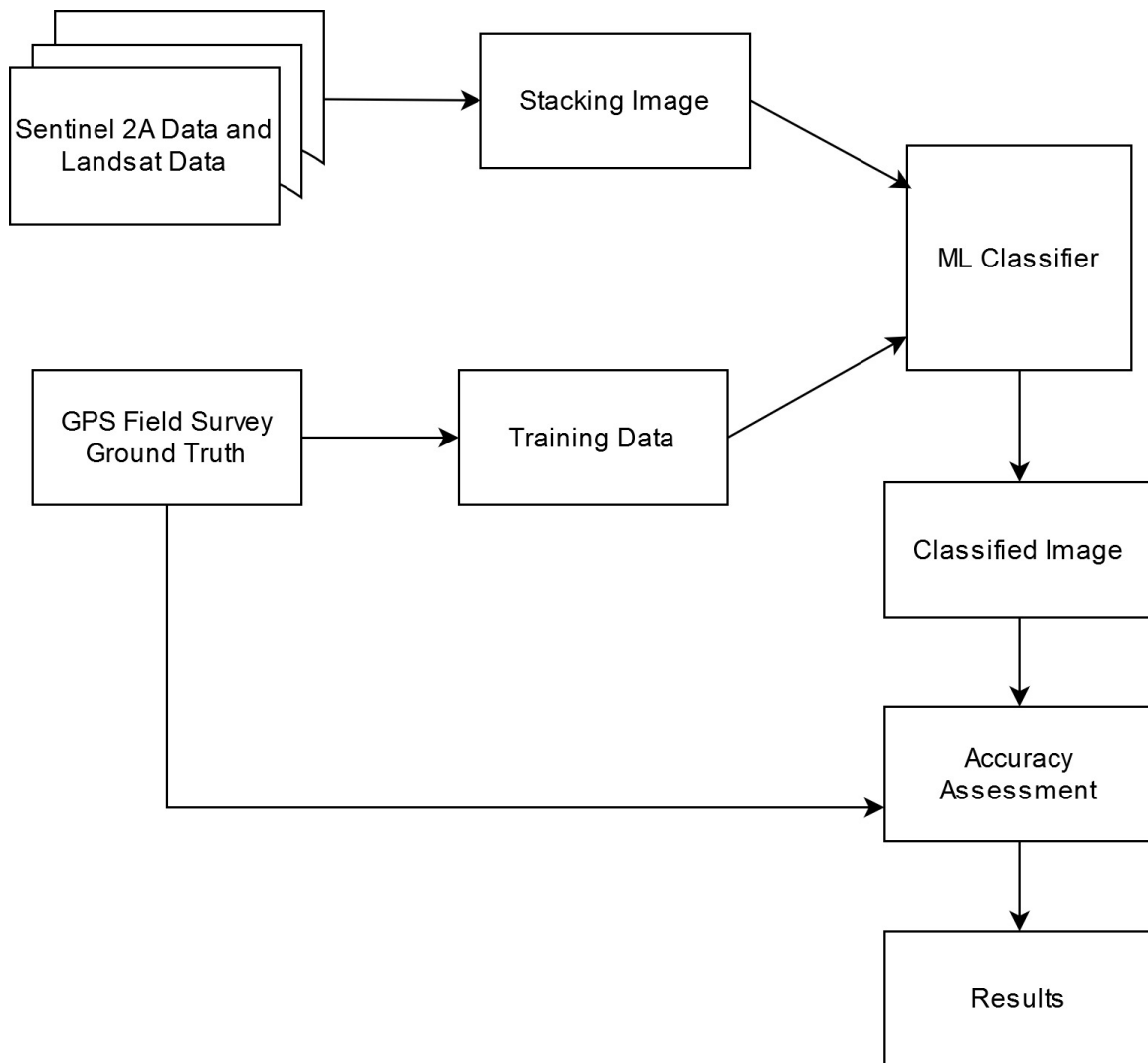


Figure 3.2: General Architecture

3.3 Methodology

In this project we are going to use a false colour image obtained by stacking different images captured by different spectral bands of the satellite, here we are going to use NIR, R, G and B bands, as the vegetation reflects in NIR band we would represent vegetation by red

colour whereas the water doesn't not reflect any of the radiations so it would appear black on the final stacked image this will help us to distinguish between vegetation and water. We are going to study two agricultural areas/villages located in Sangli district of Maharashtra State, the ground truth of these areas will be taken by recording the Global Positioning System (GPS) coordinates of the areas and then we will draw polygons based on the ground truth taken by the GPS on the stacked image classifying them in different categories such as vegetation (types of crops), barren land, water bodies and so on. This will provide us with the data of the fields.

The next step is to train the data using this information, once the models are trained, we will then classify the crops using classifiers such as Naive Bayes, Support Vector Machine (SVM), Random Forest (RF), etc. and compare the accuracy of every classifier later on.

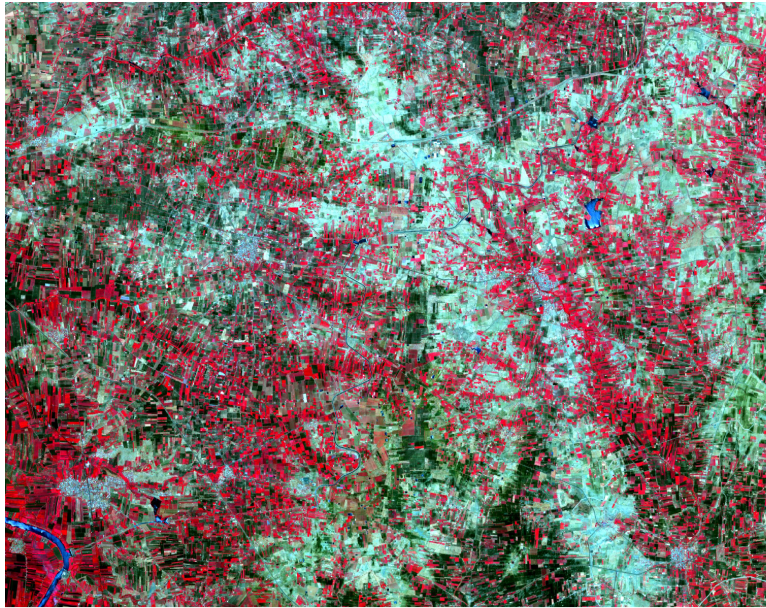


Figure 3.3: Stacked False colour image

The total number sample pixels extracted under the polygons (representing different classes) overlaid over the crop image were 17,056. Of these, a total of 10,000 samples were randomly selected for training and testing. The 10,000 samples were split in the ratio 70:30 for training and testing. That is, 70 % of the 10,000 samples were used for training while the remaining were used for testing. Table 3.1 displays total sample used for training and testing in each

class.

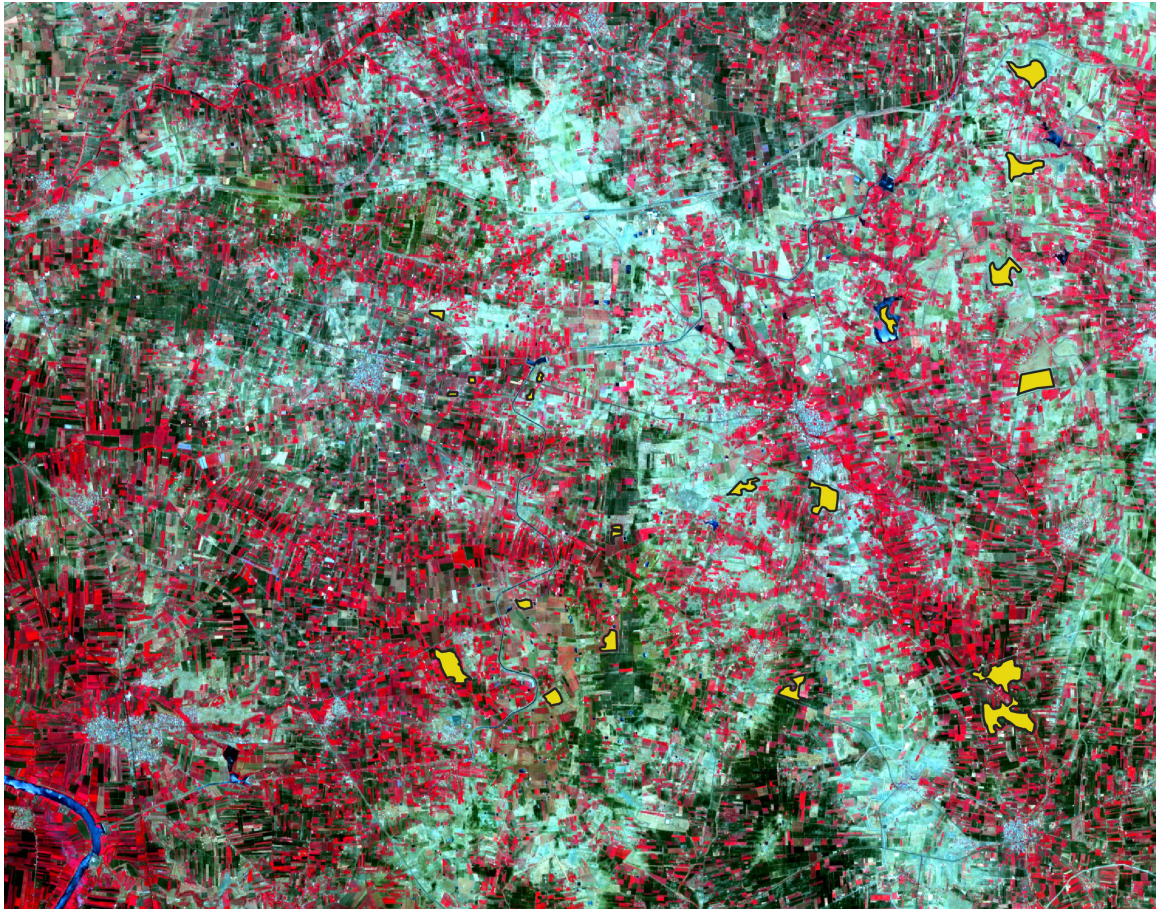


Figure 3.4: Polygons (yellow) drawn over the fields

3.4 Classification using Naive Bayes

After performing the training and testing process, the overall accuracy for Naive Bayes classifier was 97.32 %. The confusion matrix for this classifier is as shown in Table 3.2. The kappa factor was 0.9602. The statistics by class is as depicted in Table 3.3. The pixel classification using the trained Naive Bayes model is as depicted in Fig. 3.4. The colour code and the class indices represent 1 - Barren Land, 2 - Water Body, 3 - No Crop, 4 - Sugarcane, and 5 - Grapes.

Table 3.1: Training and Testing data sets used for classification

	Barren land (1)	Water body (2)	No crop (3)	Sugarcane (4)	Grapes (5)
Training samples	4675	1200	4728	265	1072
Testing samples	2035	525	1983	108	465

Table 3.2: Confusion Matrix of the Naive Bayes Classifier

	Reference				
Prediction	1	2	3	4	5
1	2023	9	0	0	3
2	10	505	3	0	0
3	0	11	1926	36	6
4	0	0	26	69	0
5	2	0	28	3	456

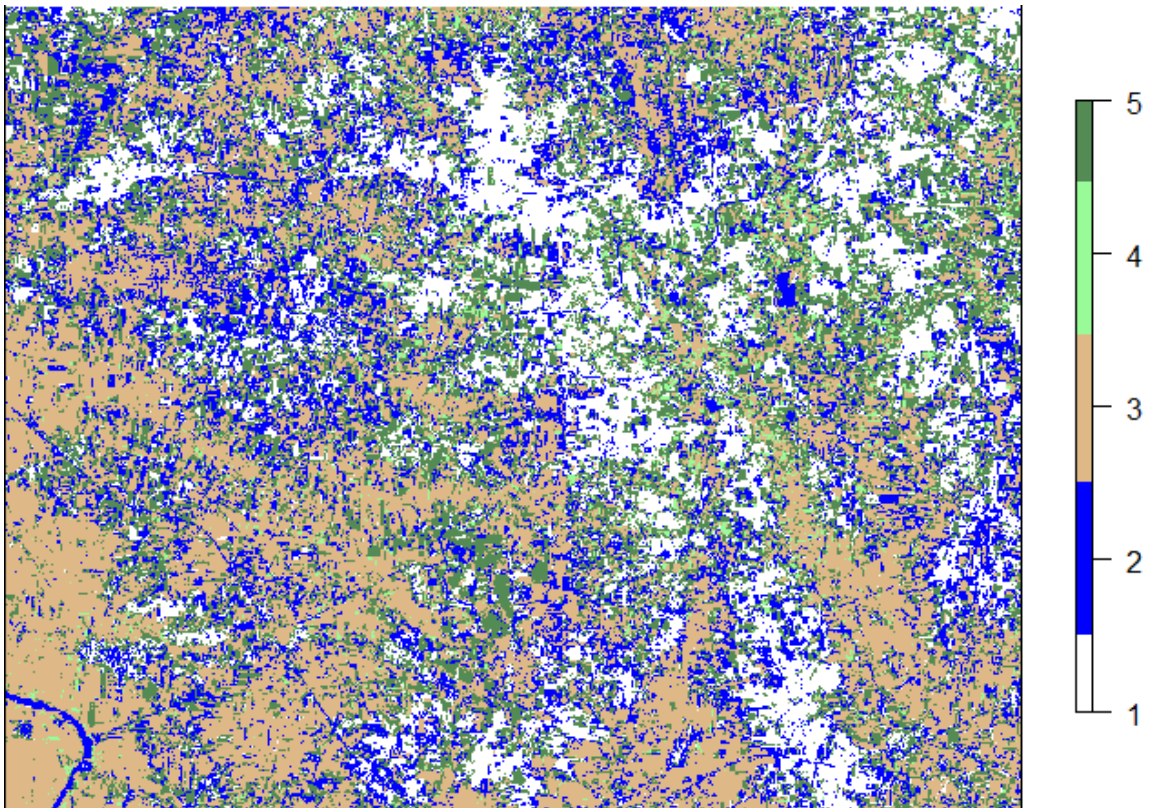


Figure 3.5: Classification using Naive Bayes

Table 3.3: Statistics by Class for Naive Bayes classifier

	Class: 1	Class: 2	Class: 3	Class: 4	Class: 5
Sensitivity	0.9941	0.96190	0.9713	0.63889	0.98065
Specificity	0.9961	0.99717	0.9831	0.99481	0.99290
Positive Predicted Value	0.9941	0.97490	0.9732	0.72632	0.93252
Negative Predicted Value	0.9961	0.99565	0.9818	0.99223	0.99805
Prevalence	0.3978	0.10262	0.3876	0.02111	0.09089
Detection Rate	0.3954	0.09871	0.3765	0.01349	0.08913
Detection Prevalence	0.3978	0.10125	0.3868	0.01857	0.09558
Accuracy	0.9951	0.97954	0.9772	0.81685	0.98677

3.5 Classification using SVM

After performing the training and testing process, the overall accuracy for the SVM classifier was 98.26 %. The confusion matrix for this classifier is as shown in Table 3.4. The kappa factor was 0.9741. The statistics by class is as depicted in Table 3.5.

Table 3.4: Confusion Matrix of the SVM Classifier

	Reference				
Prediction	1	2	3	4	5
1	2032	0	0	0	2
2	0	523	0	0	0
3	0	2	1944	33	5
4	0	0	14	72	2
5	3	0	25	3	456

The pixel classification using the trained SVM model is as depicted in Fig. 3.5. The colour code and the class indices represent 1 - Barren Land, 2 - Water Body, 3 - No Crop, 4 - Sugarcane, and 5 - Grapes.

Table 3.5: Statistics by Class for SVM classifier

	Class: 1	Class: 2	Class: 3	Class: 4	Class: 5
Sensitivity	0.9985	0.9962	0.9803	0.66667	0.98065
Specificity	0.9994	1.0000	0.9872	0.99681	0.99333
Positive Predicted Value	0.9990	1.0000	0.9798	0.81818	0.93634
Negative Predicted Value	0.9990	0.9996	0.9875	0.99284	0.99806
Prevalence	0.3978	0.1026	0.3876	0.02111	0.09089
Detection Rate	0.3972	0.1022	0.3800	0.01407	0.08913
Detection Prevalence	0.3976	0.1022	0.3878	0.01720	0.09519
Accuracy	0.9989	0.9981	0.9838	0.83174	0.98699

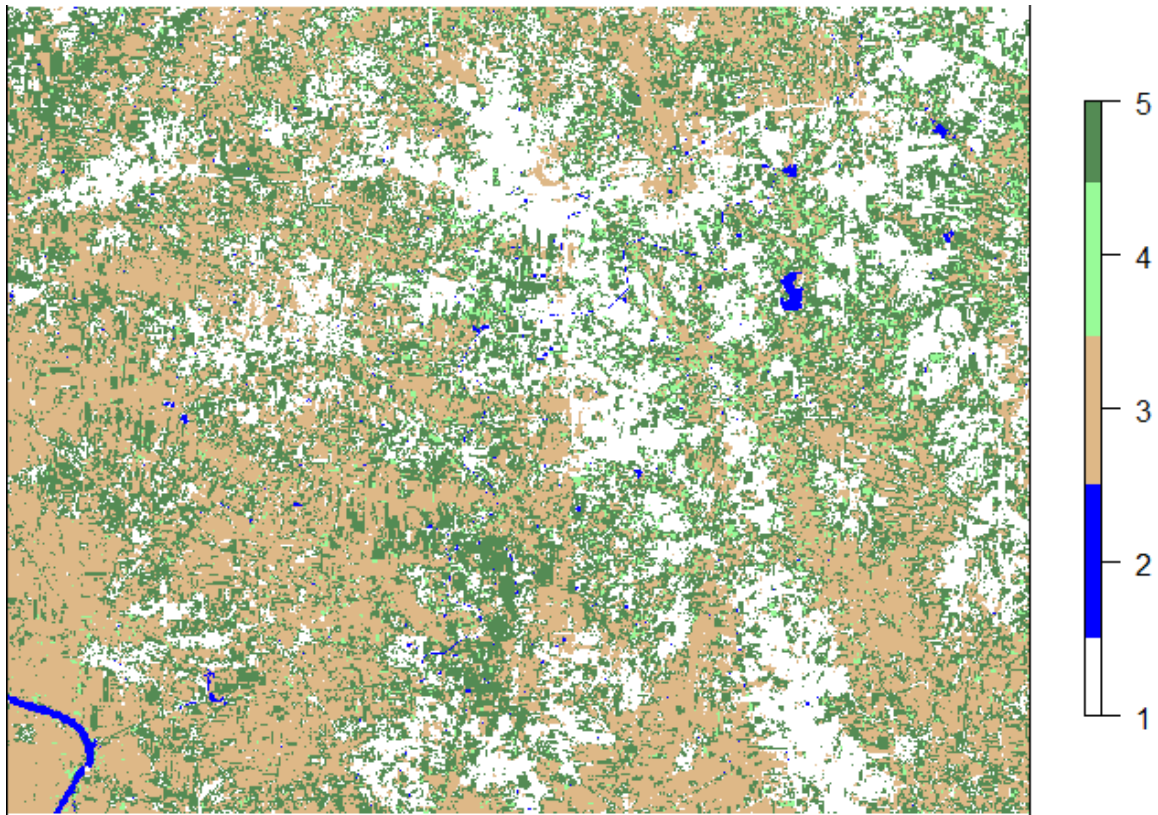


Figure 3.6: Classification using SVM

3.6 Classification using Random forests

After performing the training and testing process using the Random forests classifier, the overall accuracy observed was 98.59 %. The confusion matrix for this classifier is as shown in Table 3.6. The kappa factor was 0.9791. The statistics by class is as depicted in Table 3.7.

Table 3.6: Confusion Matrix of the Random forests Classifier

	Reference				
Prediction	1	2	3	4	5
1	2031	0	0	0	2
2	1	521	0	0	0
3	0	4	1947	21	4
4	0	0	17	86	0
5	3	0	19	1	459

Table 3.7: Statistics by Class for Random forests classifier

	Class: 1	Class: 2	Class: 3	Class: 4	Class: 5
Sensitivity	0.9980	0.9924	0.9818	0.79630	0.98710
Specificity	0.9994	0.9998	0.9907	0.99661	0.99505
Positive Predicted Value	0.9990	0.9981	0.9853	0.83495	0.95228
Negative Predicted Value	0.9987	0.9991	0.9885	0.99561	0.99871
Prevalence	0.3978	0.1026	0.3876	0.02111	0.09089
Detection Rate	0.3970	0.1018	0.3806	0.01681	0.08972
Detection Prevalence	0.3974	0.1020	0.3862	0.02013	0.09421
Accuracy	0.9987	0.9961	0.9863	0.89645	0.99108

The pixel classification using the trained Random forests model is as depicted in Fig. 3.6. The colour code and the class indices represent 1 - Barren Land, 2 - Water Body, 3 - No Crop, 4 - Sugarcane, and 5 - Grapes.

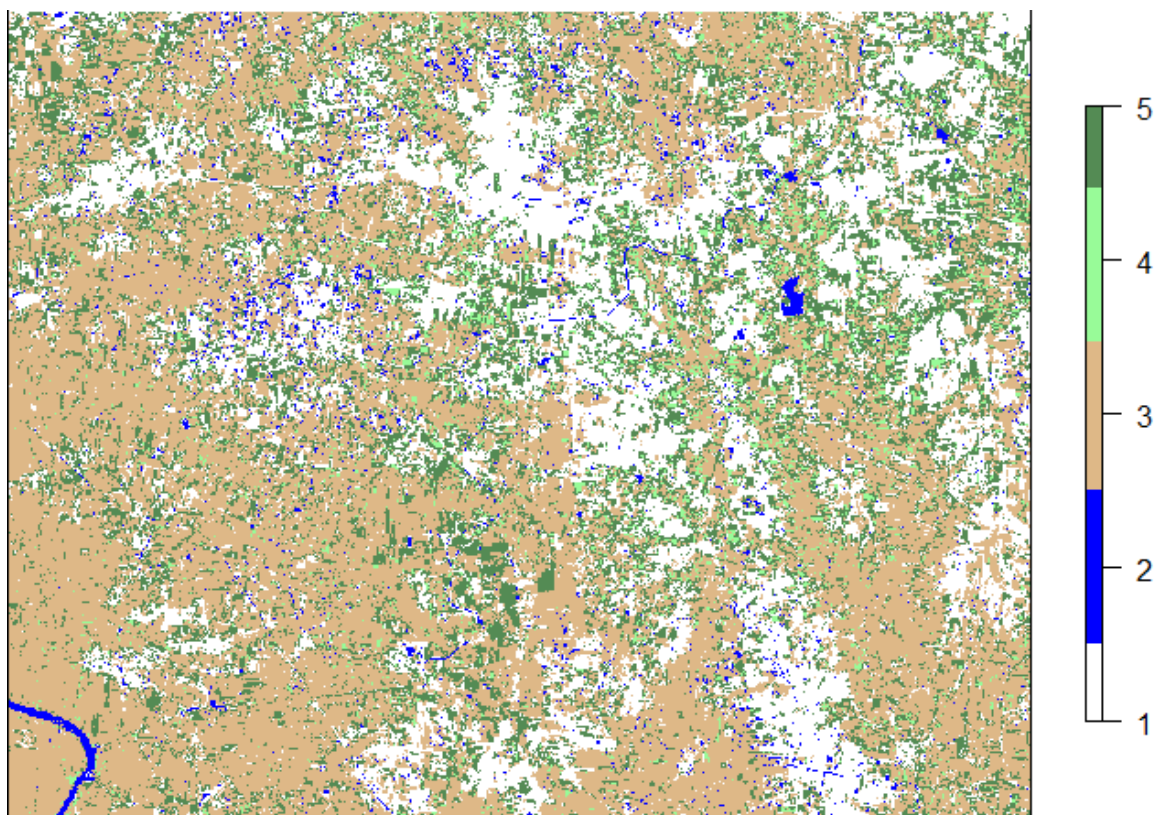


Figure 3.7: Classification using Random forests

3.7 Classification using Uniform samples

Since the number of ground truth samples in the sugarcane class (class 4) is less in comparison with other classes, we consider the complete set of pixels values for training and testing, that is, 17,056 sample pixels. These 17,056 sample pixels were split in the ratio 70:30 for training and testing. That is, 70 % of the 17,056 samples were used for training while the remaining were used for testing. Table 3.8 displays total sample used for training and testing in each class.

Finally, 250 sample pixels from training set of each class were randomly selected for uniform training. Similarly, 100 sample pixels from testing set of each class were randomly selected

Table 3.8: Training and Testing data samples from complete data set

	Barren land (1)	Water body (2)	No crop (3)	Sugarcane (4)	Grapes (5)
Training samples	4653	1193	4757	261	1076
Testing samples	2057	532	1954	112	461

and used for testing. The number of samples used for uniform training and testing are as shown in Table 3.9.

Table 3.9: Training and Testing data samples for uniform training and testing

	Barren land (1)	Water body (2)	No crop (3)	Sugarcane (4)	Grapes (5)
Training samples	250	250	250	250	250
Testing samples	100	100	100	100	100

After performing the training and testing process, the overall accuracy after using uniform sampling for each class and the Random forests classifier was 96.8 %. The confusion matrix for this classifier is as shown in Table 3.10. The kappa factor was 0.9791. The statistics by class is as depicted in Table 3.11.

Table 3.10: Confusion Matrix using uniform sampling and Random forests Classifier

	Reference				
Prediction	1	2	3	4	5
1	99	0	0	0	1
2	0	100	0	1	0
3	0	0	93	2	3
4	0	0	3	96	0
5	1	0	4	1	96

The pixel classification using uniform samples for training and testing for each class and the corresponding trained Random forests model is as depicted in Fig. 3.7. The colour code and the class indices represent 1 - Barren Land, 2 - Water Body, 3 - No Crop, 4 - Sugarcane, and 5 - Grapes.

Table 3.11: Statistics by Class for uniform sampling with Random forests classification

	Class: 1	Class: 2	Class: 3	Class: 4	Class: 5
Sensitivity	0.9900	1.0000	0.9300	0.9600	0.9600
Specificity	0.9975	0.9975	0.9875	0.9925	0.9850
Positive Predicted Value	0.9900	0.9901	0.9490	0.9697	0.9412
Negative Predicted Value	0.9975	1.0000	0.9826	0.9900	0.9899
Prevalence	0.2000	0.2000	0.2000	0.2000	0.2000
Detection Rate	0.1980	0.2000	0.1860	0.1920	0.1920
Detection Prevalence	0.2000	0.2020	0.1960	0.1980	0.2040
Accuracy	0.9938	0.9988	0.9587	0.9763	0.9725

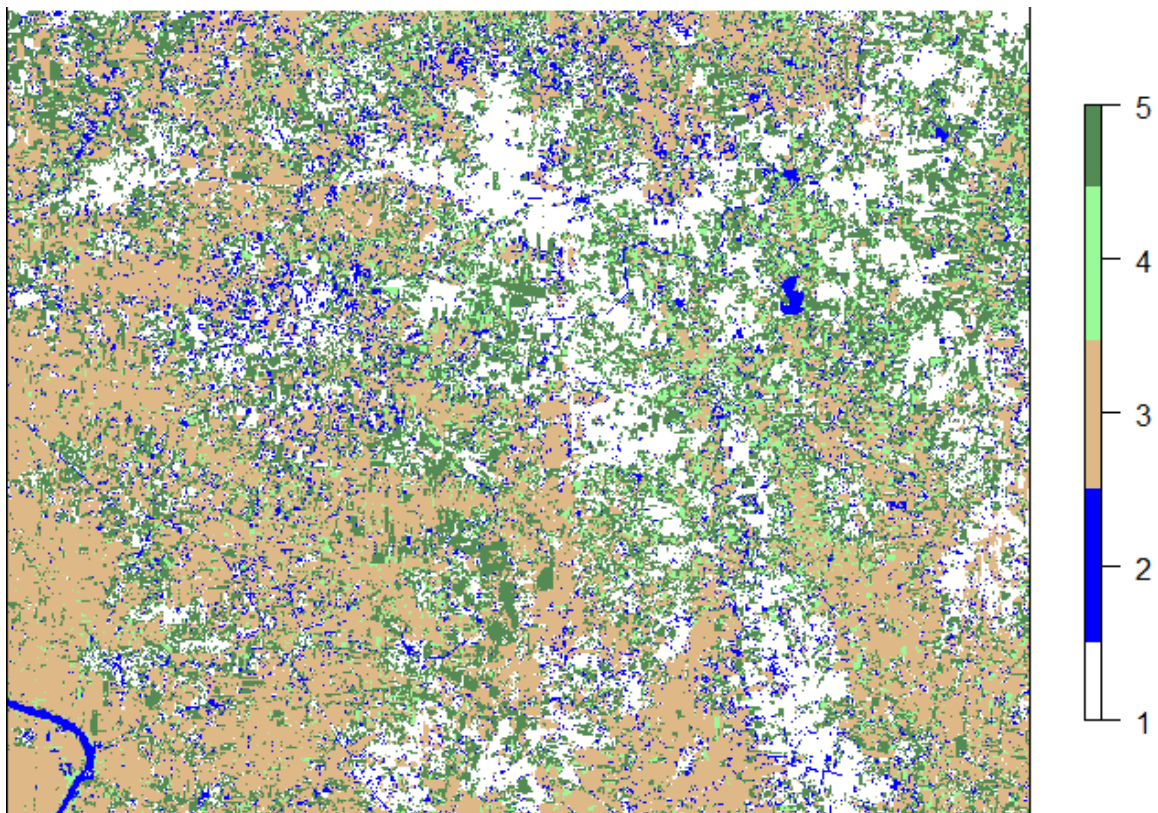


Figure 3.8: Classification using Uniform samples and Random forests

Chapter 4

Results and Discussion

Crop classification and crop yield have become important aspects of modern agriculture. They help in better understanding of different crops cultivated across the globe and growth of crops during various stages of their lifecycle. Some popular crop categories (and sub-categories) include food crops (cereals, pulses, fruits, vegetables, etc.), industrial crops (cotton, sugarcane, tobacco, castor, groundnut, etc.), fodder crops, etc. With the advancements in satellite technology and computing power, automated crop classification and yield calculation has gained significant research interest in recent times.

The aim of this dissertation is to use different machine learning algorithms to perform automated crop classification. In this study we have used Sentinel-2 satellite data (10 metre resolution; Apr 10, 2017) of agricultural crop areas in Arag and Bedag villages in Sangali District, Maharashtra. Various GPS points within the crop areas were collected and mapped to corresponding pixel coordinates in the satellite image data. These pixel coordinates were used to draw the polygons depicting the crop and non-crop classes, thereby giving the ground truth pixels for the study. Three classification algorithms have been in the study, namely, Nave Bayes, SVM and Random forests. Amongst various bands in the Sentinel-2 data, we have used three bands (NIR, Red, and Green) for classification. The NIR band was the most useful band as vegetation reflects the radiation that was necessary for the classification of the crops.

4.1 Classification results using different classifiers

The implementation details of the crop classification have been discussed in Chapter 3. The overall accuracy using confusion matrix and the kappa factor are popularly used for evaluation of classification accuracy. Also, tuning of parameters for various classifiers such as SVM and Random forests directly impacts the classification accuracy. For SVM, the penalty parameter (C) and Gamma (γ) need to be estimated and appropriately set to get better accuracy. In our study C was estimated as 70 and Gamma (γ) was set to 1. In case of Random forests the n_trees parameter representing number of trees and m_try parameter representing the number of predictors need to appropriately set to get better classification results. The optimal value for n_trees was 400 and m_try was set to 1 in our study.

The Nave Bayes classifier is the most conventional classifier whose accuracy depends on the availability of good ground truth crop pixel values. The overall accuracy for this classifier was 97.32 %, while kappa factor value was 0.9602. The SVM classifier gave an accuracy of 98.26 %, with a kappa factor value of 0.9741. The Random forests classifier has the highest accuracy in comparison to other classifiers. The Random forest classifier accuracy was 98.59% and its kappa factor value was 0.9791. All the classifiers also classify other entities such as barren land, water bodies, urban sprawl, etc. The Random forest classifier is an ensemble classifier and hence, as expected, it has a better classification accuracy.

4.2 Classification result using uniform sampling

The overall classification accuracy can sometimes be misleading, especially when the training samples are not uniform across the classes. For example, sugarcane class (Class 4) has less number of ground truth samples in comparison with other classes. In section 3.7, we have performed uniform sampling of different crop categories (and other classes) and performed the training. We have used Random forest classifier for training and testing. The overall classification accuracy observed after uniform sampling is 96.8 % (Kappa factor value 0.9791). Although the overall accuracy comes down, the accuracy across all the class is uniform. For example, the overall accuracy for sugarcane class using the Random forest classifier was 0.89645 (Table 3.7), which improved to 0.9767 (Table 3.10) using uniform

sampling. Also, it can be seen that there is less mixing of pixels across the classes in uniform sampling.

Chapter 5

Conclusion

Crop classification involves three entities, namely; data, model (classifier), and the experts who provide the ground truth. In this project we have used 10-metre resolution Sentinel-2 data spawning Arag and Bedag villages in Sangali District, Maharashtra. The ground truth data comprising GPS points and the corresponding crops associated with each GPS point was collected manually. The GPS points were later mapped to pixel positions in the image to create polygons depicting each crop (and non-crop) category. The classification algorithms used in the study comprised Nave Bayes, SVM and Random forests. Amongst various bands in the Sentinel-2 data, we have used three bands (NIR, Red, and Green) for classification. The NIR band was the most useful band as vegetation reflects the radiation that was necessary for the classification of the crops.

5.1 Summary of the results

The results and implementation details of the crop classification have been discussed in Chapter 3. The overall accuracy for Nave Bayes classifier was 97.32 %, while the SVM classifier gave an accuracy of 98.26 %, and the Random forest classifier gave an accuracy of 98.59%. It is observed that Random forests classifier has the highest accuracy in comparison to other classifiers for crop classification. The Random forest classifier is an ensemble classifier and therefore gives a better classification accuracy.

In order to avoid bias in classification accuracy due to skewness in number of available training samples in each class, we have carried out uniform sampling of different crop categories and performed the training (3.7). In this case we have used Random forest classifier for training and testing. The classification accuracy observed after uniform sampling is 96.8%. Although the overall accuracy comes down, the accuracy across all the class is uniform as seen in Table 3.10.

5.2 Future work and trends

Crop classification is a widely researched area across the globe. Several conventional machine learning algorithms have been used to perform crop classification. The current efforts are directed towards the use of better and faster machine learning algorithms to perform crop classification more accurately and in real-time. Due to limitations of the conventional classification algorithms on big satellite data, the current and future trend is to use neural network models and deep learning techniques for classification. The accuracy of these models is highly dependent on the quality of reference data or the ground truth. The ground truth data could also be pooled through crowdsourcing. However, there is need to make sure that the crowdsourced data is correct, cleaned and made consistent before its use.

Classification of multi-temporal satellite images of large tracts of agricultural land requires good computational resources and high speed Internet for downloading the data. This can be done smoothly by using cloud environments, like Amazon or Goggle Earth Engine, where data downloading and processing time would be significantly reduced. The European Space Agency has developed a cloud infrastructure for Copernicus Data and Information Access Services Operations (DIAS). Cloud platforms would serve as solution for scaling the small area applications to large agricultural tracts. They allow implementation of parallelization in an effective way to increase the efficiency of the system. For example, LandViewer platform allows one to perform on-the-fly analysis and downloading of satellite data. With rapid changes in technology, crop classification and mapping would become faster and more efficient for large areas by making use of high resolution images. Also, the advancement in satellite technologies that can compute complex calculations and the use of machine learning models would make the crop classification task easier.

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